

SEEM3440 Operations Research II — Assignment 1

Answers

Q1(a) — PIP exhaustive enumeration (IP0: max $6.5x_1 + 5x_2$)

Candidate integer box: x_1 in $\{0,1,2,3\}$, x_2 in $\{0,1,2,3\}$ (from $2x_1 \leq 7$, $2x_2 \leq 7$). Y = feasible, N = infeasible.

x_1	x_2	Feasible?	z
0	0	N (3)	0
0	1	Y	5
0	2	Y	10
0	3	Y	15
1	0	Y	6.5
1	1	Y	11.5
1	2	Y	16.5
1	3	N (1)	21.5
2	0	Y	13
2	1	Y	18
2	2	N (1)(2)	23
2	3	N (1)(2)	28
3	0	N (2)	19.5
3	1	N (1)(2)	24.5
3	2	N (1)(2)	29.5
3	3	N (1)(2)	34.5

Optimal: $(x_1^*, x_2^*) = (2, 1)$ with $z^* = 18$.

Q1(b) — PIP Branch-and-Bound (first fractional variable, natural order)

LP relaxation optima and fathoming (incumbent z^* starts at $-\infty$).

Node	Added constraint	LP optimum	z bound	Outcome
LP0	—	(1.5, 2)	19.75	Branch x_1
LP1	$x_1 \leq 1$	(1, 2.5)	19	Branch x_2
LP2	$x_1 \geq 2$	(2, 1)	18	Integer: fathomed; $z^* = 18$
LP3	$x_1 \leq 1, x_2 \leq 2$	(1, 2)	16.5	Integer, $16.5 \leq 18$: fathomed
LP4	$x_1 \leq 1, x_2 \geq 3$	(0.5, 3)	18.25	Branch x_1
LP5	$x_1 \leq 1, x_2 \geq 3, x_1 \leq 0$	(0, 3.5)	17.5	$17.5 \leq 18$: fathomed
LP6	$x_1 \leq 1, x_2 \geq 3, x_1 \geq 1$	—	—	Infeasible: fathomed

Optimal: $(x_1^*, x_2^*) = (2, 1)$ with $z^* = 18$. B&B tree: LP0 $\rightarrow$ {LP1, LP2}; LP1 $\rightarrow$ {LP3, LP4}; LP4 $\rightarrow$ {LP5, LP6}.

Q2(a) — BIP exhaustive enumeration (32 binary points)

Constraints: (1) $x_1 + 2x_2 - 3x_4 - x_5 \leq 0$; (2) $-15x_1 + 30x_2 - 35x_3 + 45x_4 + 45x_5 \geq 50$. Y = feasible.

$(x_1, x_2, x_3, x_4, x_5)$	z	Feas?
0,0,0,0,0	0	N
0,0,0,0,1	-1	N
0,0,0,1,0	-2	N
0,0,0,1,1	-3	Y

(x1,x2,x3,x4,x5)	z	Feas?
0,0,1,0,0	5	N
0,0,1,0,1	4	N
0,0,1,1,0	3	N
0,0,1,1,1	2	Y
0,1,0,0,0	3	N
0,1,0,0,1	2	N
0,1,0,1,0	1	Y
0,1,0,1,1	0	Y
0,1,1,0,0	8	N
0,1,1,0,1	7	N
0,1,1,1,0	6	N
0,1,1,1,1	5	Y
1,0,0,0,0	3	N
1,0,0,0,1	2	N
1,0,0,1,0	1	N
1,0,0,1,1	0	Y
1,0,1,0,0	8	N
1,0,1,0,1	7	N
1,0,1,1,0	6	N
1,0,1,1,1	5	N
1,1,0,0,0	6	N
1,1,0,0,1	5	N
1,1,0,1,0	4	Y
1,1,0,1,1	3	Y
1,1,1,0,0	11	N
1,1,1,0,1	10	N
1,1,1,1,0	9	N
1,1,1,1,1	8	Y

Optimal: (x1*,x2*,x3*,x4*,x5*) = (1, 1, 1, 1, 1) with z* = 8 (unique optimum; 9 feasible points).

Q2(b) — BIP Branch-and-Bound (LP-solver printouts)

LP relaxation ($0 \leq x_j \leq 1$) optima; branching on first fractional variable.

Node	Fixed vars	LP optimum	z bound	Outcome
LP0	—	(1, 1, 1, 0.7222, 0.8333)	8.7222	Branch x4
LP1	x4 = 0	(0, 0.5, 0.2857, 0, 1)	1.9286	1.9286 ≤ 8: fathomed
LP2	x4 = 1	(1, 1, 1, 1, 0.5556)	8.4444	Branch x5
LP3	x4 = 1, x5 = 0	(1, 1, 0.2857, 1, 0)	5.4286	5.4286 ≤ 8: fathomed
LP4	x4 = 1, x5 = 1	(1, 1, 1, 1, 1)	8	Integer: fathomed; z* = 8

Optimal: (1, 1, 1, 1, 1) with z* = 8. B&B tree: BIP0 → {BIP1 (x4=0), BIP2 (x4=1)}; BIP2 → {BIP3 (x5=0), BIP4 (x5=1)}.